

ECON 1550: International Finance

Money, Interest Rates, and Exchange Rates

A model of the money market

Exogenous variables		Endogenous variables			
Variable	Description	Variable	Description	Equation	Type of equation
Y	Real income	R	Domestic interest rate	$M^d/P = L(R, Y)$	Behavioral equation
M^s	Money supply	M^d	Money demand	$M^d = M^s$	Equilibrium condition
P	Price level				

Description of the Money Market

- There are only two assets: money and domestic bonds
- Money
 - Can be used for transactions
 - Pays no interest
- Bonds
 - Cannot be used for transactions
 - Pay interest $R \geq 0$

Money Demand

- Money demand is higher when:
 - Higher price level $P \rightarrow$ need more money to buy goods
 - Lower nominal interest rate $R \rightarrow$ bonds less attractive
 - Higher income $Y \rightarrow$ want more money to buy more
- Capture idea with a behavioral equation

$$M^d = P \times L\left(\underset{(-)}{R}, \underset{(+)}{Y}\right)$$

Real money demand

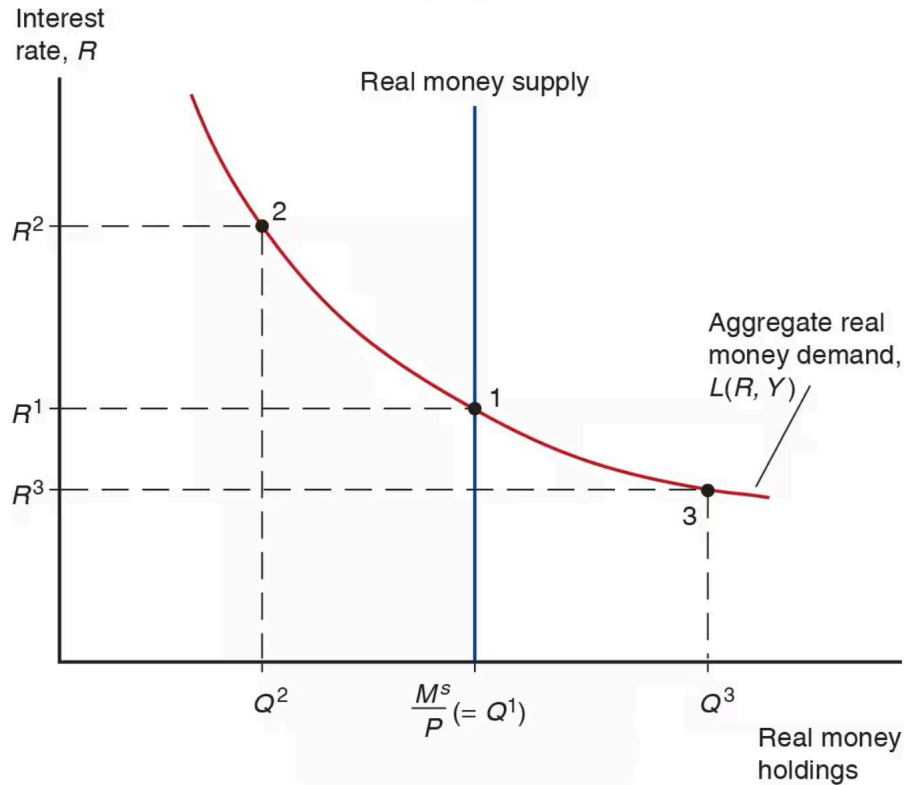
- Convenient to write in terms of real money demand

$$\frac{M^d}{P} = L(R, Y)$$

and real money supply

$$\frac{M^s}{P}$$

Equilibrium in money market



Money is defined by its function

- Medium of exchange
- Store of value
- Unit of account

Many types of money (1/2)

- Commodity money: a physical commodity (like gold) that is used as money
- Convertible paper money: a piece of paper that can be exchanged for a commodity

Many types of money (2/2)

- Fiat money: issued by a central bank, not backed by any commodity
- Digital currency: not backed by any commodity, privately issued, electronic payments

Fiat money

1. Central bank liabilities are special because they are the unit of account.
2. Currency is a promise to deliver future central bank liabilities.

Assets	Liabilities	Assets	Liabilities	Assets	Liabilities
\$0	\$0	\$1 currency	\$1 money	\$1 bonds	\$1 money
$t = 0$		$t = 1$		$t = 2$	

Monetary policy

- Deposits at the Fed are called reserves
- “The” interest rate is just interest on reserves

Short-run FX and money market model

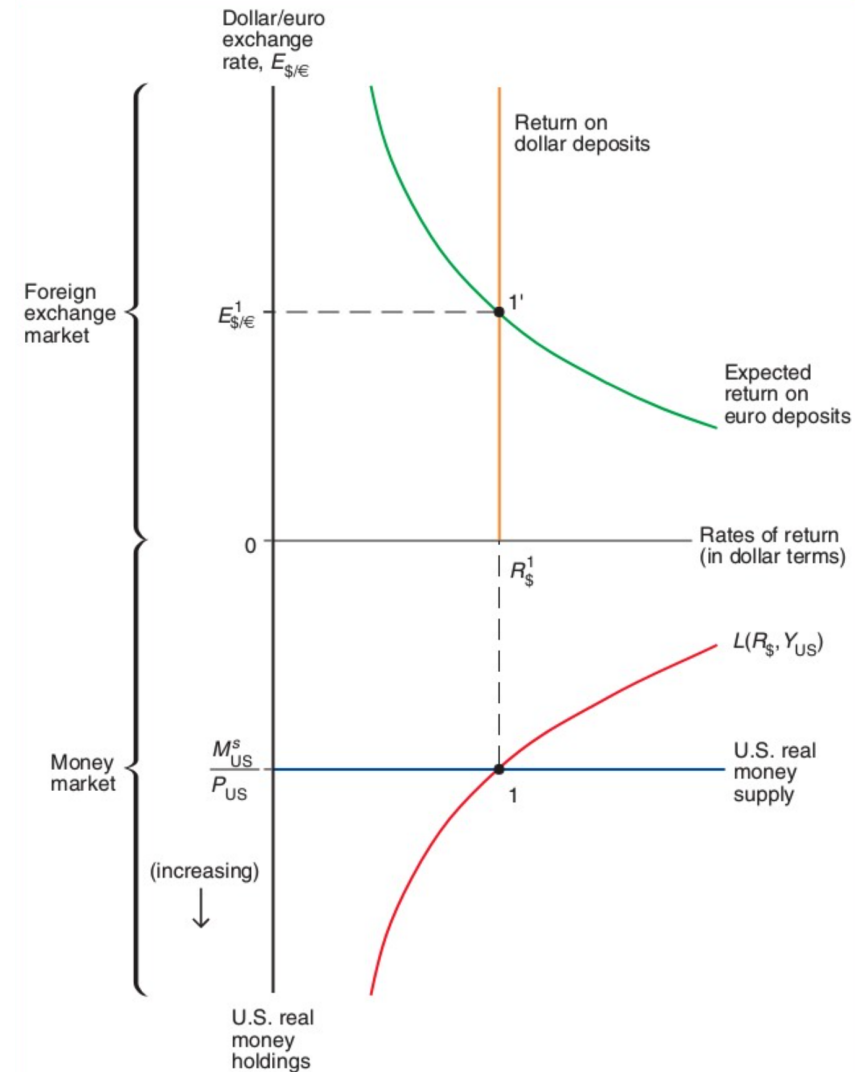
Exogenous variables

Variable	Description
R^*	Foreign interest rate
E^e	Expected exchange rate
Y	Real income
M^s	Money supply
P	Price level

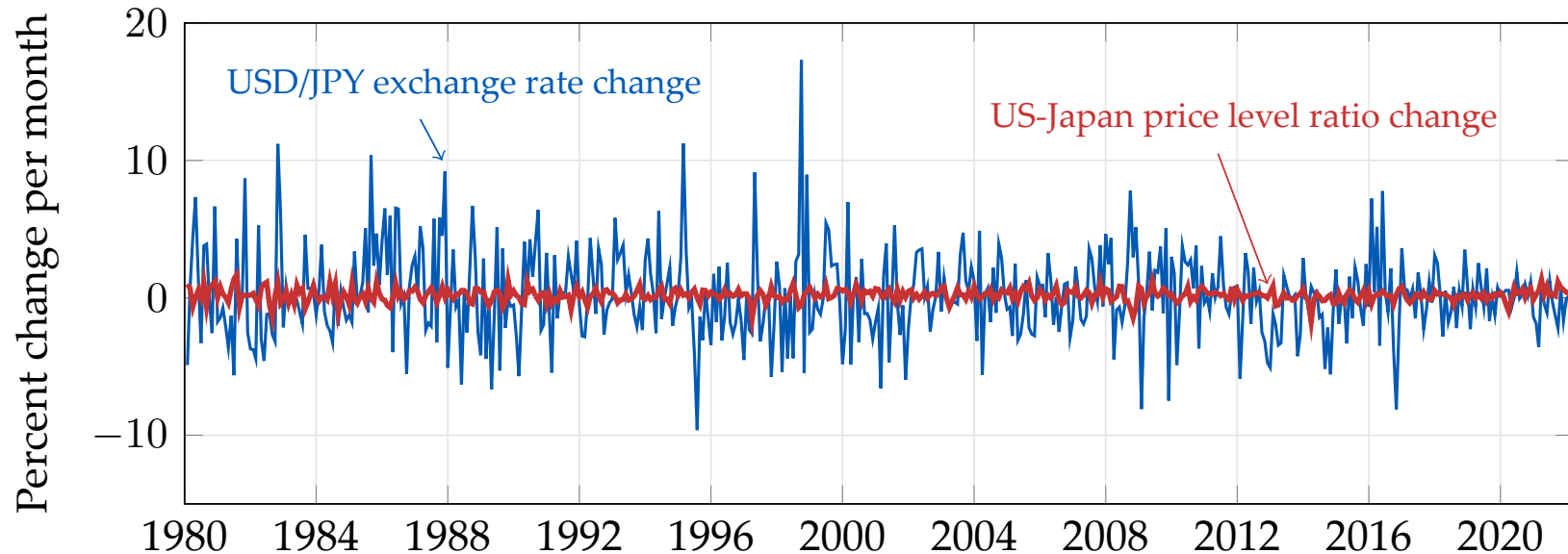
Endogenous variables

Variable	Description	Equation	Type of equation
E	Exchange rate	$R = R^* + \frac{E^e - E}{E}$	Equilibrium condition
R	Domestic interest rate	$M^d/P = L(R, Y)$	Behavioral equation
M^d	Money demand	$M^d = M^s$	Equilibrium condition

Short-run equilibrium in FX and money markets



Exchange rates are very volatile



Source: [FRED/CPIAUCSL](#); [FRED/CPALCY01JPM661N](#); [FRED/DEXJPUS](#) (end-of-month, inverted to USD/JPY).

Conceptual definition of long-run equilibrium

- Hypothetical equilibrium that would result if economy runs indefinitely with no shocks
- Equivalently, the hypothetical equilibrium that would occur if prices were perfectly flexible and always adjusted instantaneously and frictionlessly

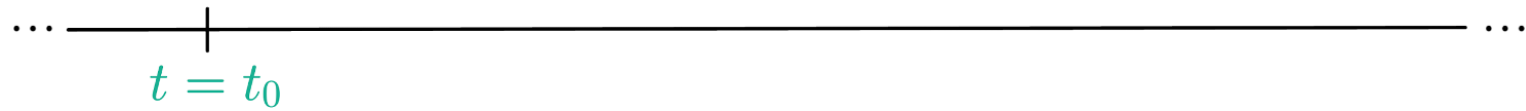
Definition of the long run in the model

1. In the long run, $E = E^e$
 - By definition of the long run, all variables constant
 - By assumption of rational expectations, expectations must be correct in the long run
 - In the model, it means $E_{LR} = E_{LR}^e$

Assumptions about the short run

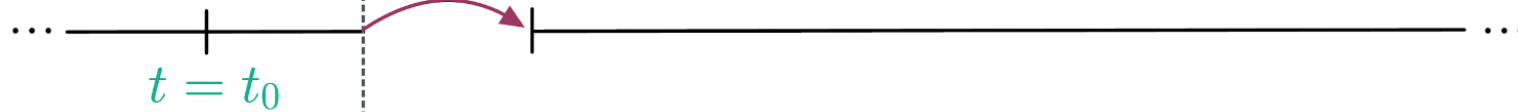
1. In the short run, P is fixed
 - Because prices are “sticky”
 - In the model, it means $P_0 = P_{SR}$
2. In the short run, E^e equals long-run E
 - By assumption of rational expectations
 - In the model, it means $E_{SR}^e = E_{LR}$

Initial long run
equilibrium



Initial long run
equilibrium

Short run



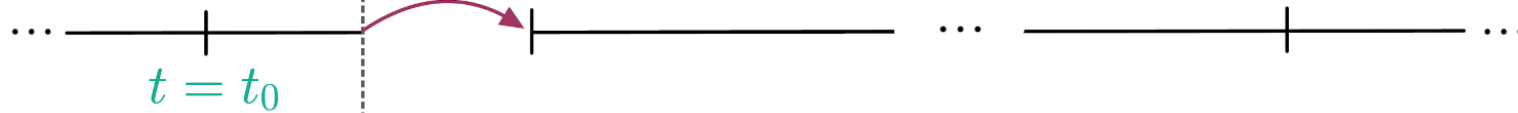
Exogenous
variable
changes

A red arrow points from the text "Exogenous variable changes" to the vertical dashed line.

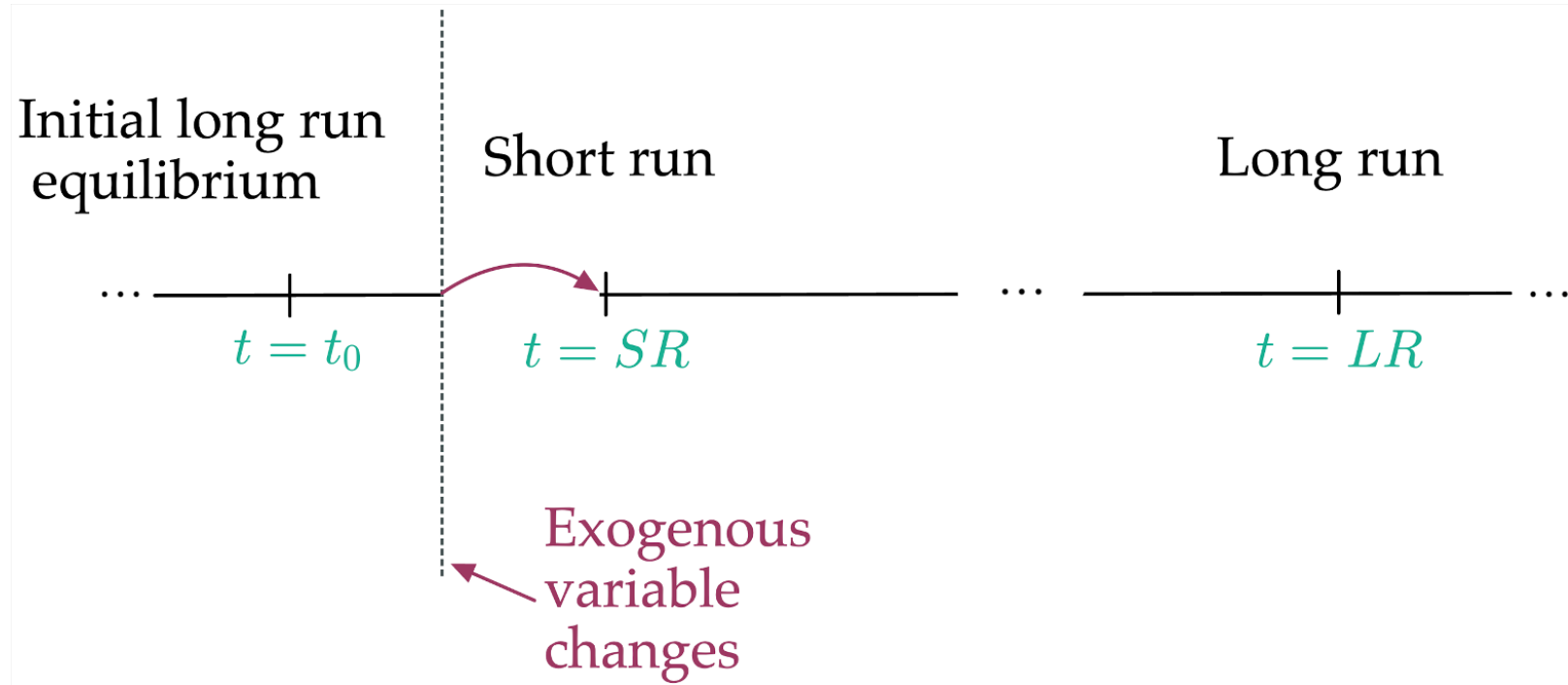
Initial long run
equilibrium

Short run

Long run



Exogenous
variable
changes



Results in the long run (1/3)

1. Long-run interest rate R_{LR} from FX market
 - Using $E_{LR} = E_{LR}^e$ and UIP:

$$R_{LR} = R^* + \frac{E_{LR}^e}{E_{LR}} - 1$$
$$\Rightarrow R_{LR} = R^*$$

Results in the long run (2/3)

2. Long-run price level P_{LR} from the money market
- Using $R_{LR} = R^*$ and money market equilibrium condition:

$$\frac{M^s}{P_{LR}} = L(R_{LR}, Y) = L(R^*, Y)$$
$$\Rightarrow P_{LR} = \frac{M^s}{L(R^*, Y)}$$

Results in the long run (3/3)

3. Long-run exchange rate E_{LR} from PPP (purchasing power parity)

- Because E_{LR} is a nominal price, in the long run it moves proportionally to the price level:

$$E_{LR} \propto P_{LR}$$

Results in the short run (1/2)

1. Short-run exchange rate E_{SR} from UIP with $E^e = E_{LR}$
 - Using UIP and the short-run assumptions:

$$R = R^* + \frac{E_{LR} - E_{SR}}{E_{SR}} \quad \Rightarrow \quad E_{SR} = \frac{E_{LR}}{1 + R - R^*}$$

Results in the short run (2/2)

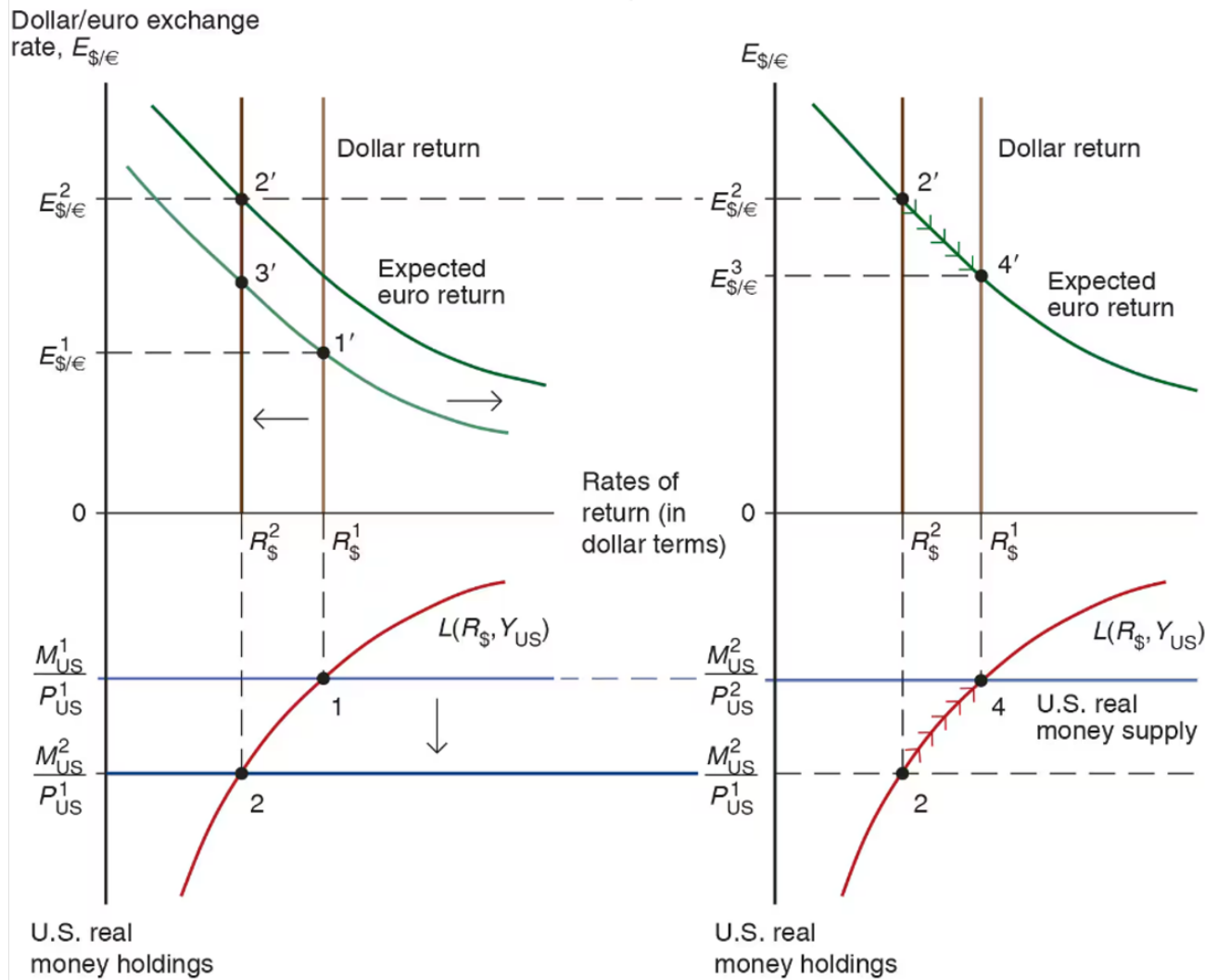
2. Short-run domestic interest rate R_{SR} from the money market

- Using money market equilibrium condition and P fixed at P_0 in the short run:

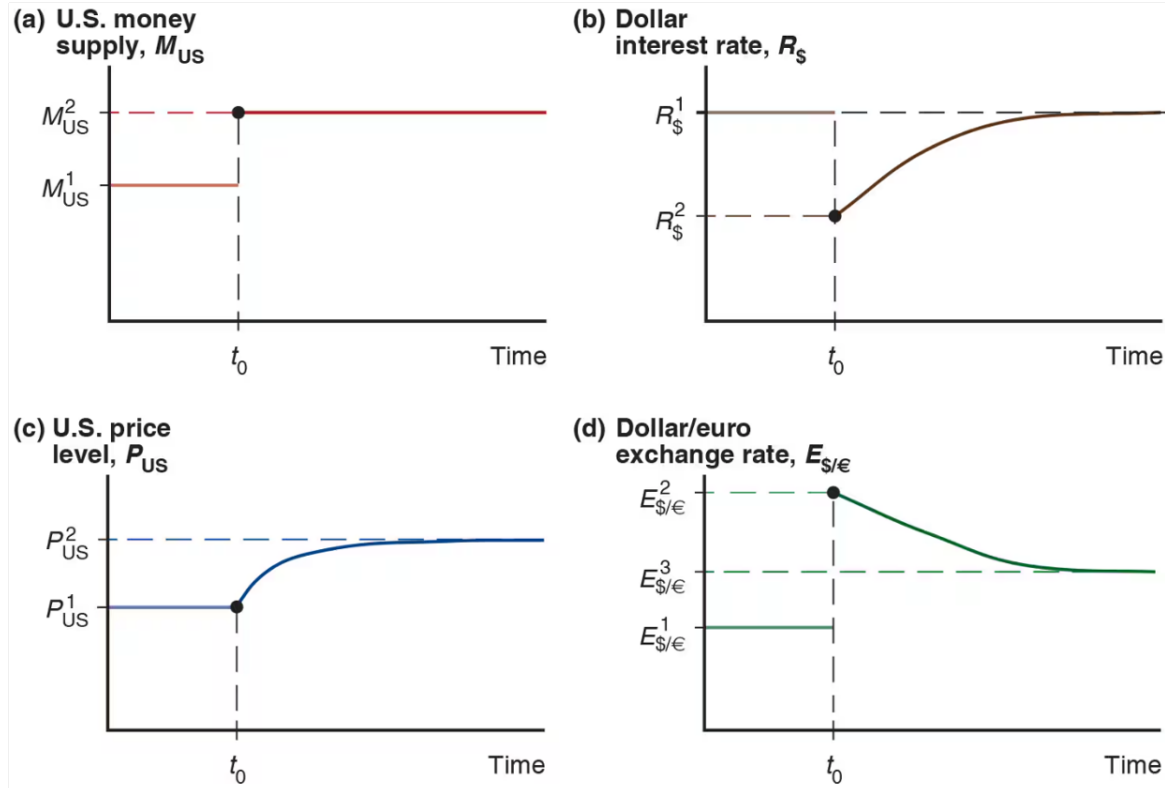
$$\frac{M^s}{P_{SR}} = L(R_{SR}, Y) \quad \Rightarrow \quad \frac{M^s}{P_0} = L(R_{SR}, Y),$$

which can be solved for R_{SR}

Permanent Money Supply Changes



Exchange Rate Overshooting



Long-run FX and money market model

Exogenous variables

Variable	Description
Y^n	Potential output
M^s	Money supply
R^*	Foreign interest rate

Endogenous variables

Variable	Description	Equation	Type of eq
E	Exchange rate	$R = R^* + \frac{E^e - E}{E}$	Eq. condition
M^d	Money demand	$M^d/P = M^s/P$	Eq. condition
E^e	Expected exchange rate	$E^e = E$	Behavioral
R	Domestic interest rate	$P = \frac{M^d}{L(R, Y^n)}$	Eq. condition
P	Price level	P fixed in the short run	Behavioral

Overshooting

1. Properties of initial, SR, LR equilibria
2. Dynamic FX and money market model
3. Example
 - Solve with equations
 - Solve graphically

Properties of initial, SR, and LR equilibria

Long run:

- $E^e = E$

$$\Rightarrow E_0^e = E_0 \text{ and } E_{LR}^e = E_{LR}$$

- E proportional to P and M^s

$$\Rightarrow E_0 = kM_0^s \text{ and } E_{LR} = kM_{LR}^s \quad (k \text{ is a parameter})$$

Properties of initial, SR, and LR equilibria

Short run:

- Prices are fixed

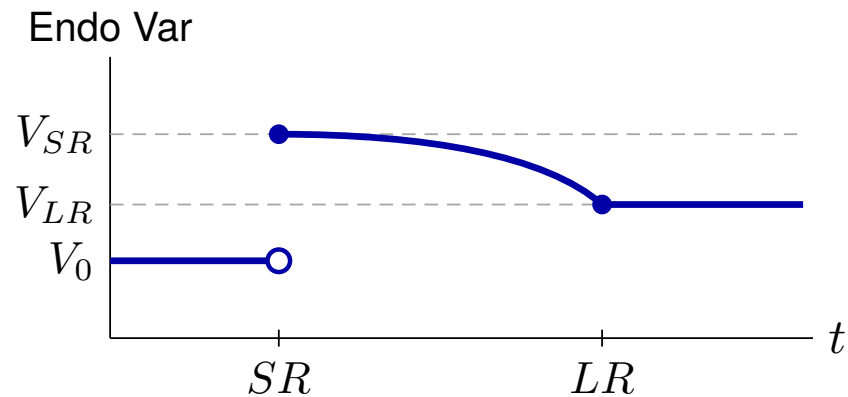
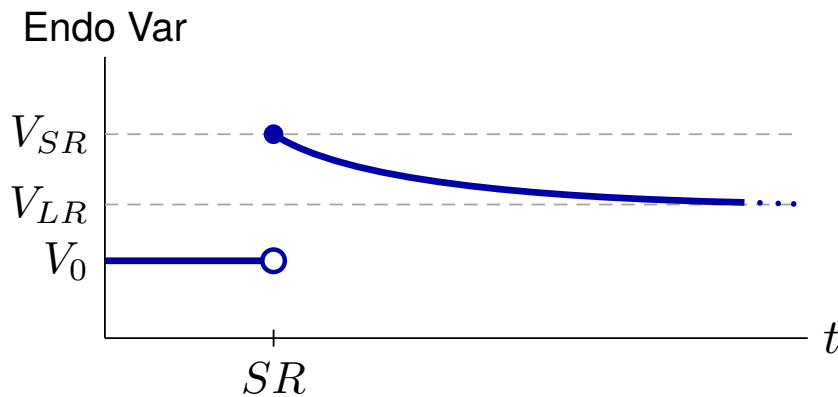
$$\Rightarrow P_{SR} = P_0$$

- Rational expectations

$$\Rightarrow E_{SR}^e = E_{LR}$$

Transition between SR and LR

- Slow monotonic transition between SR value and LR value
- No indication of shape



Dynamic FX and money market model

Exogenous variables

Variable	Description
M^s	Path of money supply
R^*	Path of foreign interest rate
Y	Path of income (GNP)
k	Long-run proportionality constant for E and M^s

Endogenous variables

Variable	Description
E	Exchange rate
E^e	Expected exchange rate
P	Price level
R	Domestic interest rate
M^d	Money demand

Equations that apply at all times

$$\text{(UIP)} : R = R^* + \frac{E^e}{E} - 1 \quad \text{(equilibrium condition)}$$

$$\text{(MD)} : \frac{M^d}{P} = L\left(R, \underset{(-)}{Y}\right) \quad \text{(behavioral)}$$

$$\text{(MS = MD)} : \frac{M^s}{P} = \frac{M^d}{P} \quad \text{(equilibrium condition)}$$

Equations that apply in SR equilibria only

(Sticky P) : $P_{SR} = P_0$

(behavioral)

(RE) : $E_{SR}^e = E_{LR}$

(behavioral)

Equations that apply in LR equilibria only

$$\text{(PPP) or (Flex P)} : E_0 = kM_0^s, \quad E_{LR} = kM_{LR}^s \quad \text{(behavioral)}$$

$$\text{(Defn LR) or (RE)} : E_0^e = E_0, \quad E_{LR}^e = E_{LR} \quad \text{(behavioral)}$$

Example

Need to specify:

1. Function $L(R, Y)$
2. Path of exogenous variables

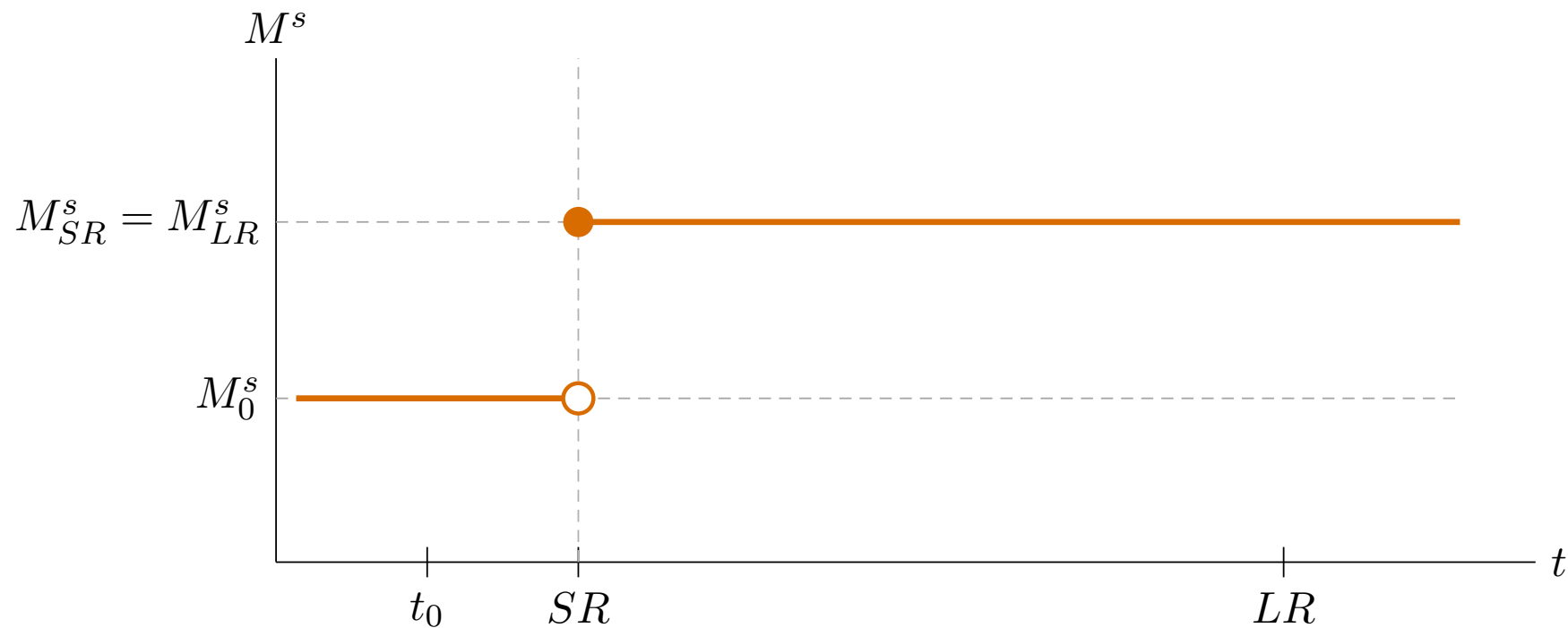
Assume:

1. $L(R, Y) = \frac{Y}{1 + R}$

2. $R^* = 0.08, Y = 1, k = 1$

$$\{M_0^s, M_{SR}^s, M_{LR}^s\} = \{1, 1.05, 1.05\}$$

Example: Permanent increase in M^s



Solution steps

1. Solve initial LR equilibrium ($t = t_0$)
2. Solve LR equilibrium ($t = LR$)
3. Solve SR equilibrium ($t = SR$)

Solution sub-steps

LR (steps 1 and 2)

- (a) Flex P / PPP $\longrightarrow E$
- (b) Defn LR / RE $\longrightarrow E^e$
- (c) UIP $\longrightarrow R$
- (d) MS = MD $\longrightarrow P$

SR (step 3)

- (a) P fixed $\longrightarrow P$
- (b) MS = MD $\longrightarrow R$
- (c) RE $\longrightarrow E^e$
- (d) UIP $\longrightarrow E$

Step 1: Solve initial LR equilibrium ($t = t_0$)

$$(a) \quad \text{Flex P / PPP} \rightarrow E \quad : \quad E_0 = kM_0^s = 1 \cdot 1 = 1$$

$$(b) \quad \text{Defn LR / RE} \rightarrow E^e \quad : \quad E_0^e = E_0 = 1$$

$$(c) \quad \text{UIP} \rightarrow R \quad : \quad R_0 = R^* + \frac{E_0^e}{E_0} - 1 \\ = 0.08 + \frac{1}{1} - 1 = 0.08$$

Step 1: Solve initial LR equilibrium ($t = t_0$)

$$\begin{aligned} \text{(d) } MS = MD \rightarrow P & \quad : \quad \frac{M_0^s}{P_0} = \frac{Y_0}{1 + R_0} \\ & \Rightarrow P_0 = M_0^s \cdot \frac{1 + R_0}{Y_0} \\ & \quad = 1 \times 1.08 = 1.08 \end{aligned}$$

Solution for t_0 : $P_0 = 1.08$, $R_0 = 0.08$, $E_0^e = 1$, $E_0 = 1$

Step 2: Solve LR equilibrium ($t = LR$)

$$(a) \quad \text{Flex P / PPP} \rightarrow E \quad : \quad E_{LR} = kM_{LR}^s = 1 \cdot 1.05 = 1.05$$

$$(b) \quad \text{Defn LR / RE} \rightarrow E^e \quad : \quad E_{LR}^e = E_{LR} = 1.05$$

$$(c) \quad \text{UIP} \rightarrow R \quad : \quad R_{LR} = R_{LR}^* + \frac{E_{LR}^e}{E_{LR}} - 1$$
$$= 0.08 + \frac{1.05}{1.05} - 1 = 0.08$$

Step 2: Solve LR equilibrium ($t = LR$)

$$\begin{aligned} \text{(d) MS=MD} \rightarrow P & \quad : \quad \frac{M_{LR}^s}{P_{LR}} = \frac{Y_{LR}}{1 + R_{LR}} \\ & \Rightarrow P_{LR} = M_{LR}^s \cdot \frac{1 + R_{LR}}{Y_{LR}} \\ & = 1.05 \times 1.08 = 1.134 \end{aligned}$$

Solution for LR: $P_{LR} = 1.134$, $R_{LR} = 0.08$, $E_{LR}^e = 1.05$, $E_{LR} = 1.05$

Step 3: Solve SR equilibrium ($t = SR$)

$$(a) \quad P \text{ fixed} \rightarrow P \quad : \quad P_{SR} = P_0 = 1.08$$

$$(b) \quad MS=MD \rightarrow R \quad : \quad \frac{M_{SR}^s}{P_{SR}} = \frac{Y_{SR}}{1 + R_{SR}}$$
$$\Rightarrow \frac{1.05}{1.08} = \frac{1}{1 + R_{SR}}$$
$$\Rightarrow 1 + R_{SR} = \frac{1.08}{1.05}$$
$$\Rightarrow R_{SR} = \frac{1.08}{1.05} - 1 \approx 0.02857$$

Step 3: Solve SR equilibrium ($t = SR$)

$$(c) \quad RE \rightarrow E^e : \quad E_{SR}^e = E_{LR} = 1.05$$

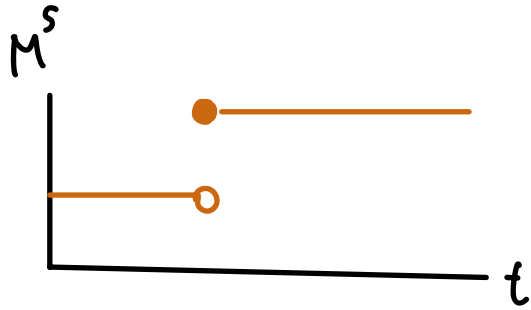
$$(d) \quad UIP \rightarrow E : \quad R_{SR} = R_{SR}^* + \frac{E_{SR}^e}{E_{SR}} - 1$$

$$\Rightarrow 0.02857 = 0.08 + \frac{1.05}{E_{SR}} - 1$$

$$\Rightarrow E_{SR} = \frac{1.05}{0.02857 - 0.08 + 1} = \frac{1.05}{0.94857} \approx 1.1069$$

Solution for SR: $P_{SR} = 1.08$, $R_{SR} \approx 0.029$, $E_{SR}^e = 1.05$, $E_{SR} \approx 1.107$

Example: solution summary

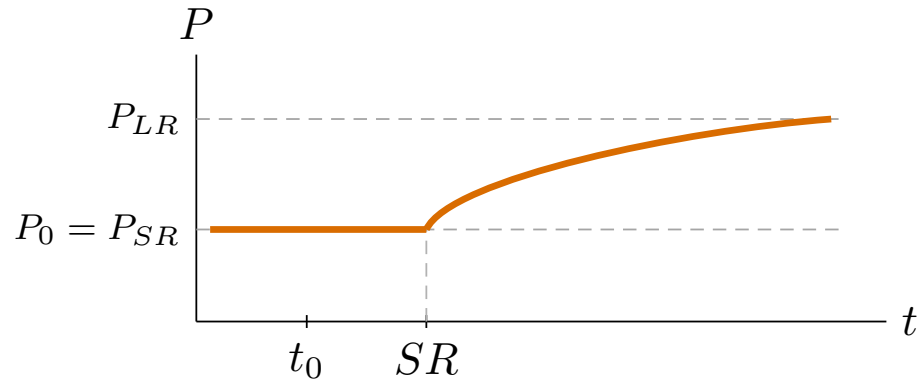
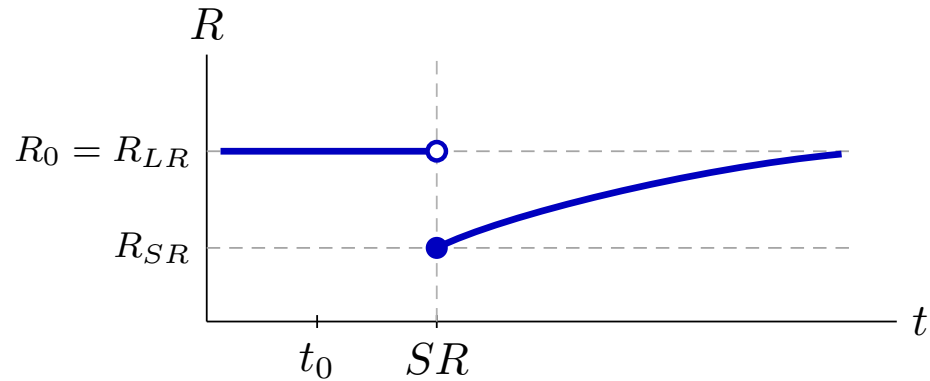
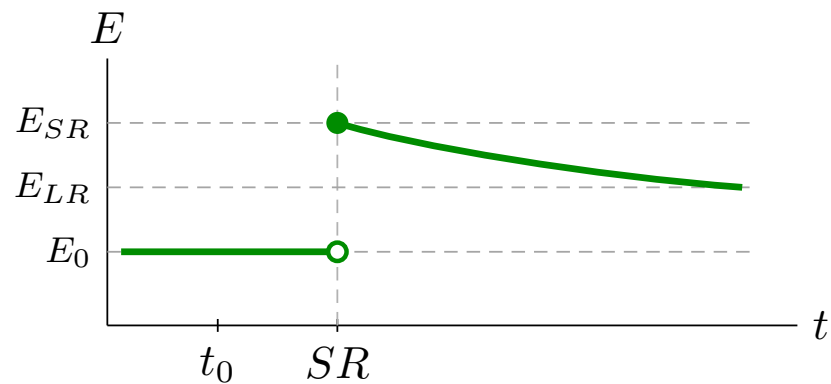
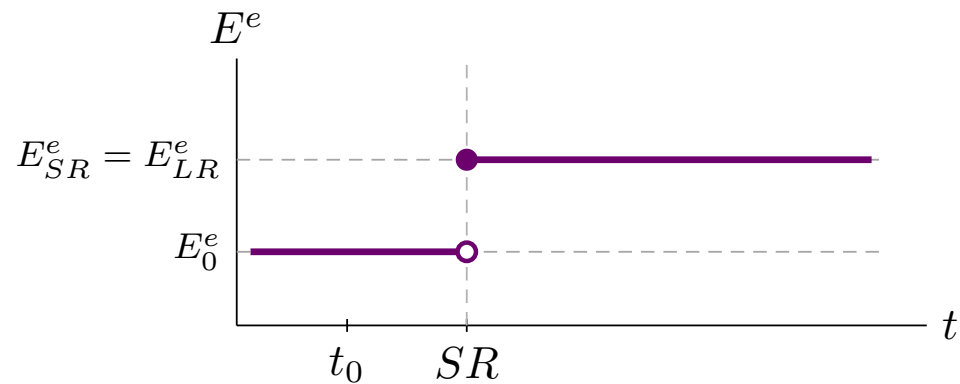


	t_0	SR	LR
P	1.08	1.08	1.134
R	0.08	0.029	0.08
E^e	1	1.05	1.05
E	1	1.107	1.05

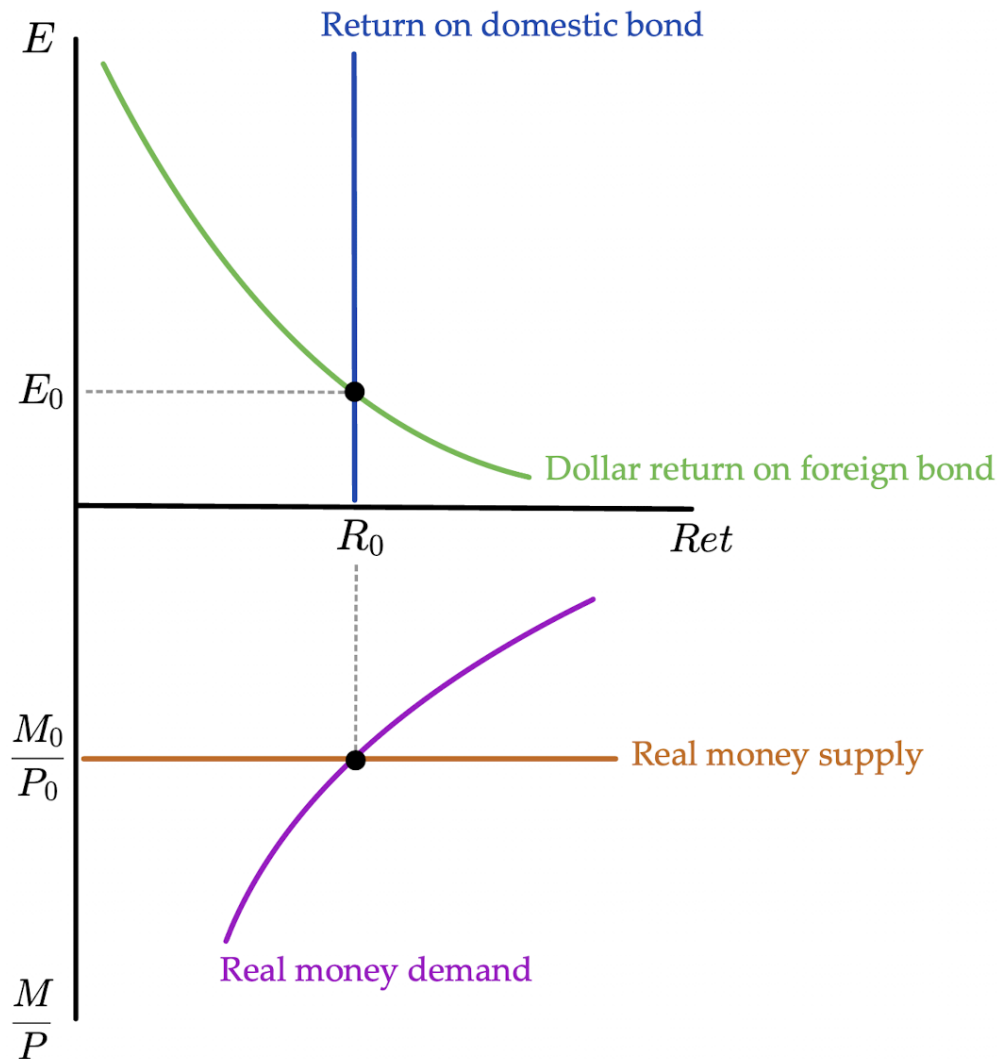
$1.107 > 1.05$
 $E_{SR} > E_{LR}$

} OVERSHOOT

Paths of solution



Solve with plots:
Initial LR
equilibrium

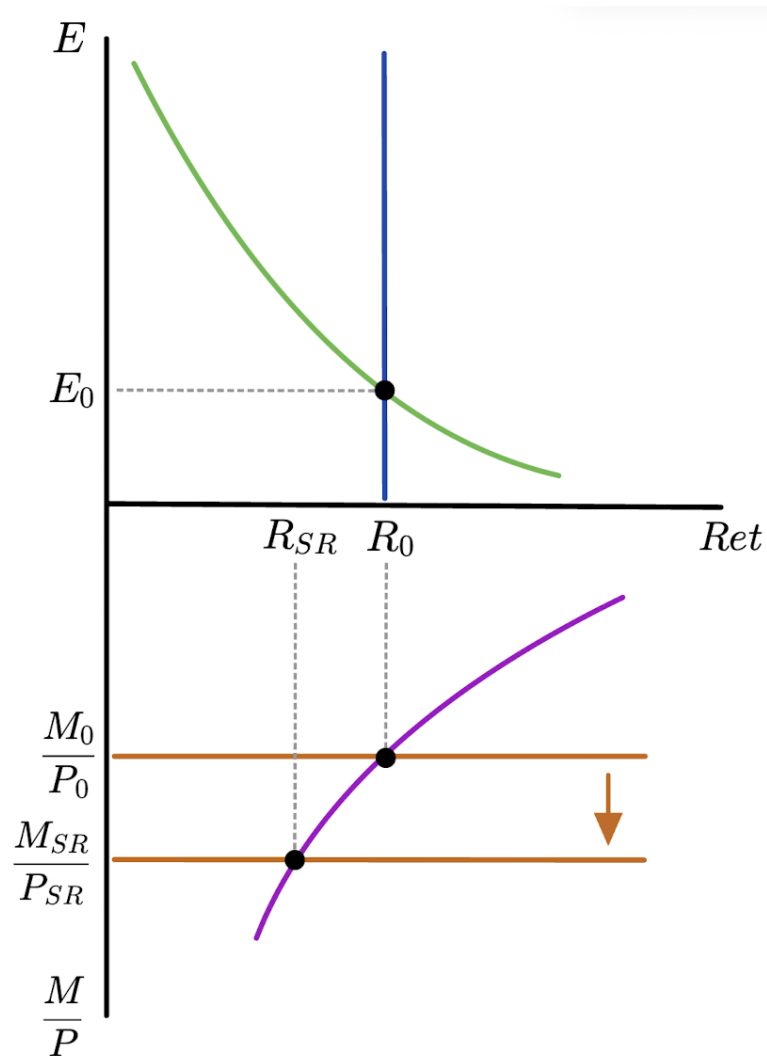


Higher M^s
lowers R

$$M^s \uparrow$$

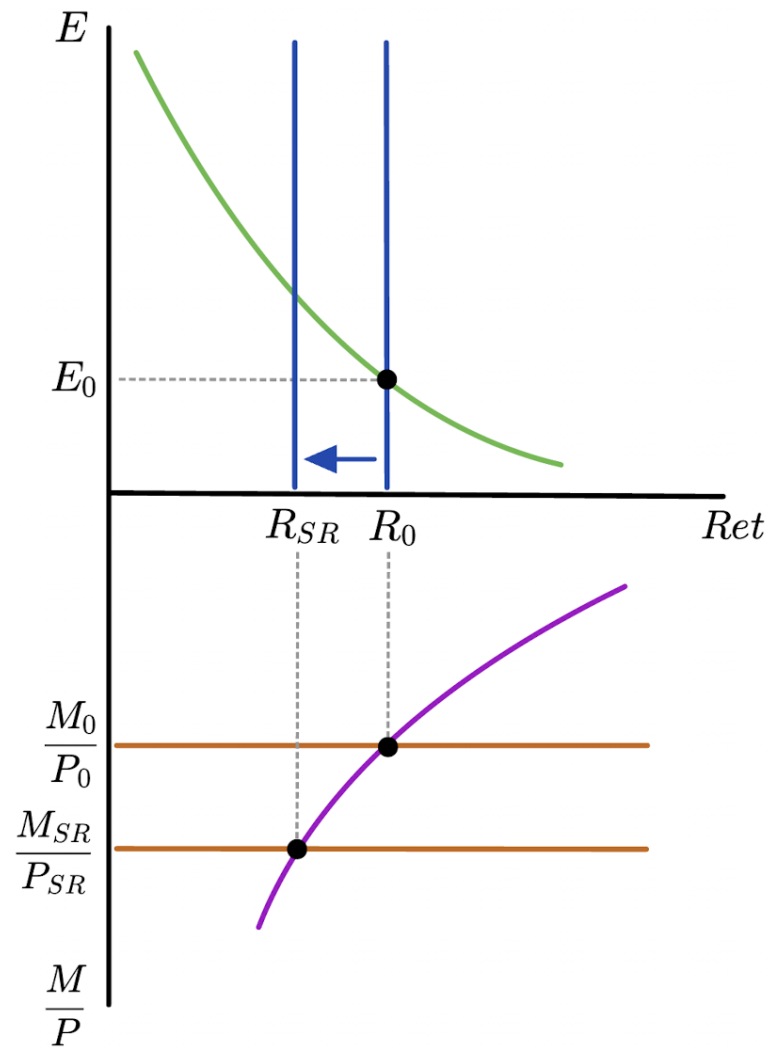
$$\frac{M^s}{P} \uparrow$$

$$R_{SR} \downarrow$$



Lower R
causes
depreciation

$R \uparrow$



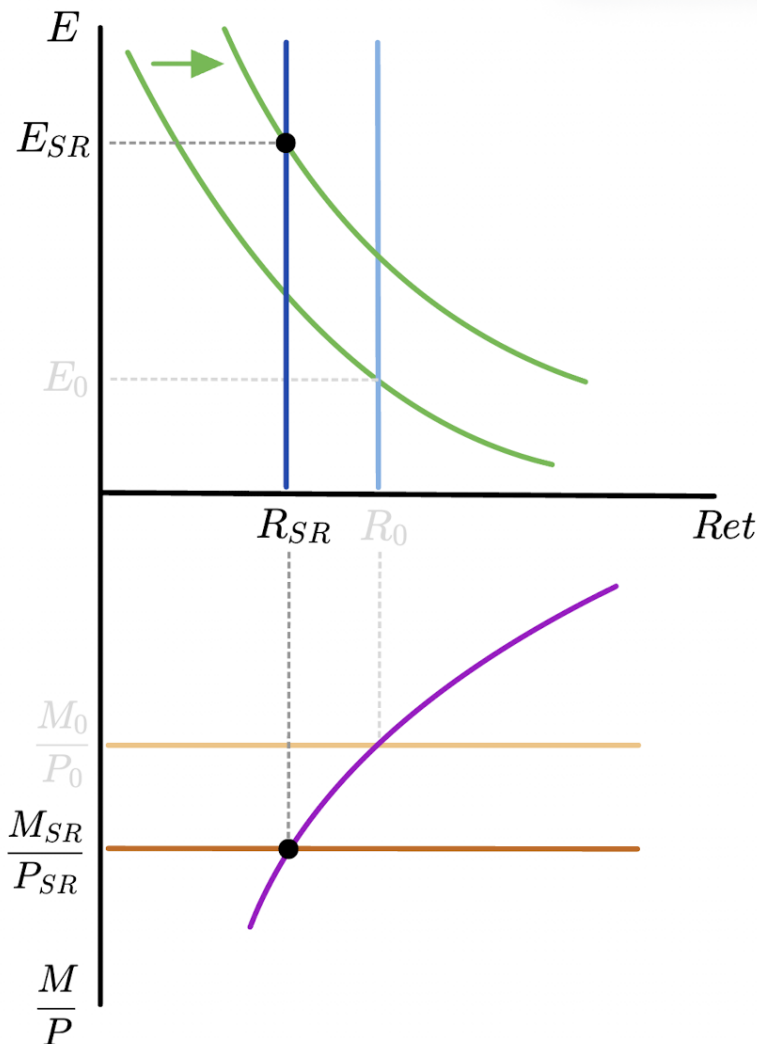
Higher E^e
shifts dollar
return on
foreign bond

$M^s \uparrow$ PERMANENTLY

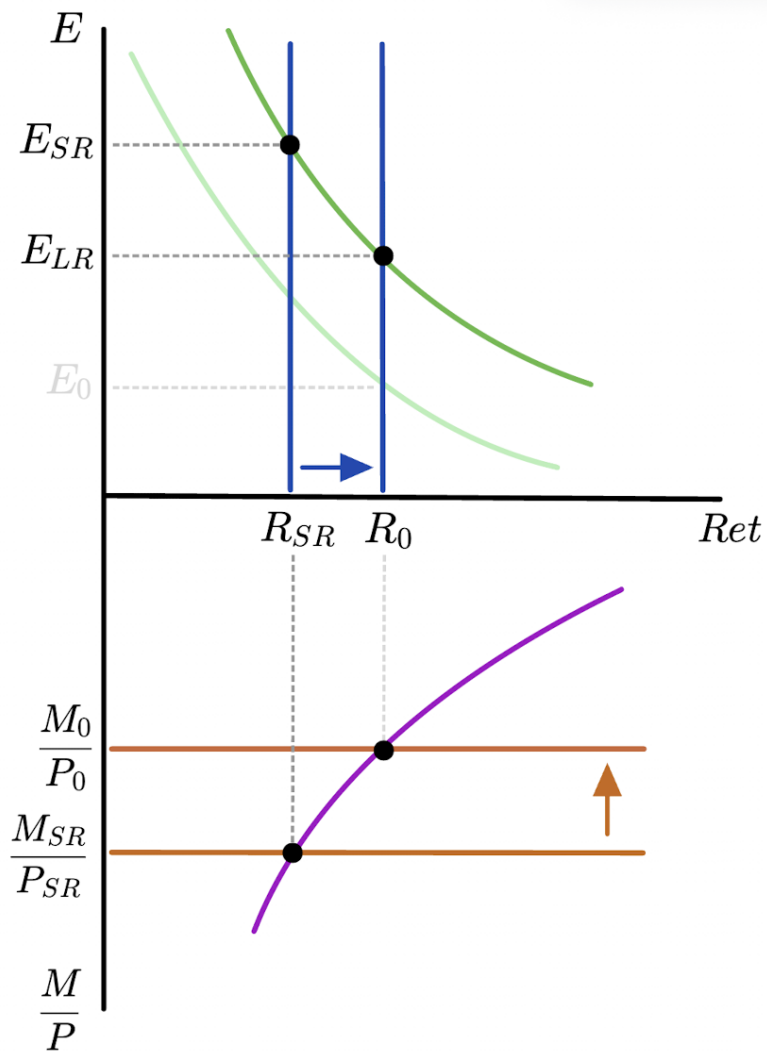
$\Rightarrow P \uparrow$ IN LR

$E \uparrow$ IN LR

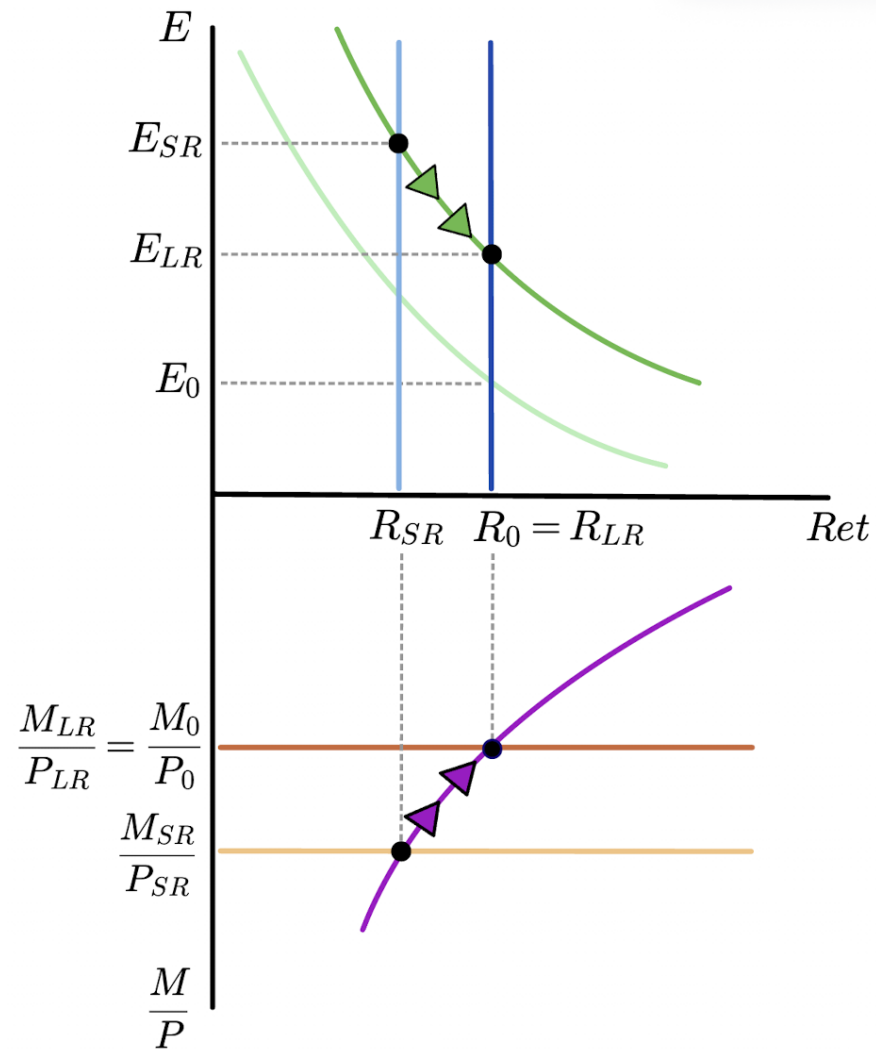
$E_{SR}^e \uparrow$



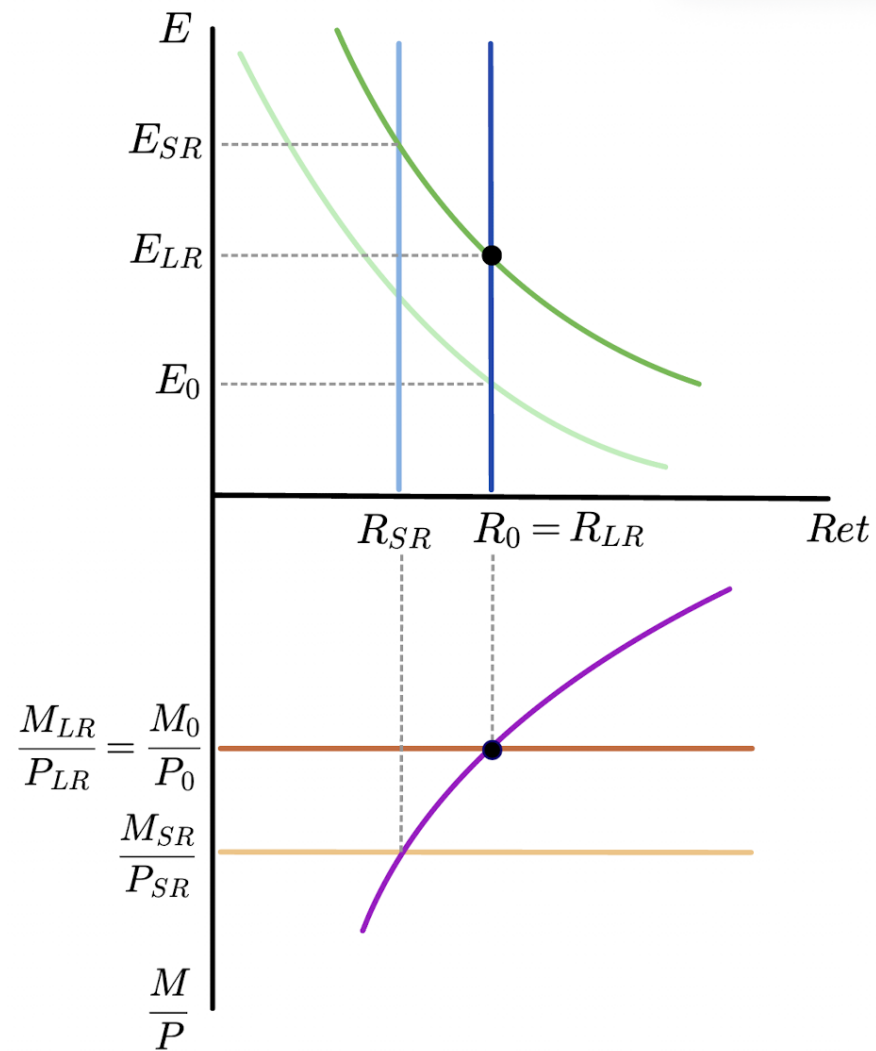
SR equilibrium



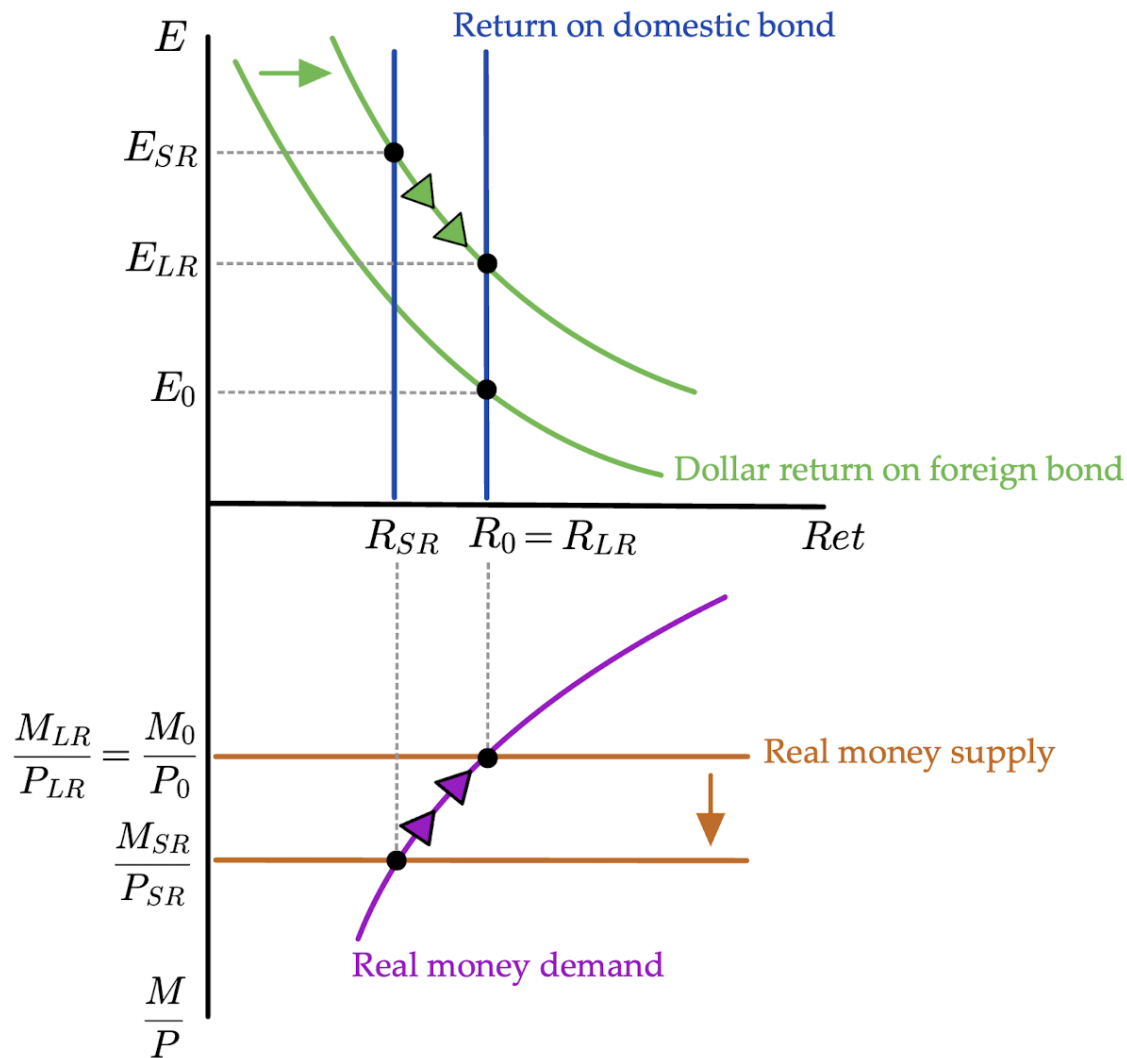
Transition between SR and LR



Final LR
equilibrium



Solve with plots:
All steps
together



Equilibrium Determination Map

	Initial	SR	LR
E	flex P / PPP	FX mkt	flex P / PPP
E^e	defn LR	RE	defn LR
R	FX mkt	money mkt	FX mkt
P	money mkt	fixed P	money mkt

PPP = purchasing power parity $\Rightarrow E \propto P \propto M^s$

defn LR = definition of the long run $\Rightarrow E_0^e = E_0$ and $E_{LR}^e = E_{LR}$

RE = rational expectations $\Rightarrow \begin{cases} E_{SR}^e = E_{LR} \\ E_0^e = E_0 \text{ and } E_{LR}^e = E_{LR} \end{cases}$